

RAJALAKSHMI ENGINEERING COLLEGE
DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE
AD23722– Project Phase-1 (2023– 2027 BATCH)

IV YEAR PROJECT - ABSTARCT (2026-27)

TITLE	Self-Refine Guardrail (SRG): A Two-Gate Framework for Automated Response Safety and Quality Verification
TEAM MEMBERS REG NO. & NAME	Jashareen J (231801066) - Meenakshi S S (231801100) – Vidhursh Kumar V (231801186)
MENTOR NAME	Mrs Rajalakshmi Elumalai
PROJECT ID	2027_PR_61

ABSTRACT:

The increasing deployment of Generative AI in customer-support systems introduces risks involving policy violations, factual inconsistencies, inappropriate communication, and confidentiality exposure. This paper presents **Self-Refine Guardrail (SRG)**, a multi-stage automated review layer designed to evaluate AI-generated responses before customer delivery. SRG employs a Review Agent for policy compliance, communication tone, and response completeness, followed by an Error-Checking Assistant responsible for factual consistency and sensitive-information detection. A deterministic Boolean routing mechanism approves responses only when both review stages satisfy their respective conditions. The study specifically investigates whether architectural separation between review stages guarantees **reasoning-level scope separation**. An initial eight-scenario evaluation achieved correct routing for all tested cases; however, overlapping policy information revealed that the Review Agent could reason about concerns assigned to the downstream stage. A five-round diagnostic progressively applied exclusion instructions, stronger rules, Boolean verdicts, visible redaction, and seamless redaction. Under the final configuration, all 10 trials were correctly blocked, while confidentiality-flavored reasoning still appeared in 9 of 10 runs. These findings indicate that automated AI review is feasible, but formal architectural boundaries do not necessarily constrain model reasoning. The study therefore identifies reasoning-level scope leakage as an important limitation requiring broader evaluation before production deployment. The results also motivate systematic testing across diverse scenarios, adversarial prompts, multi-turn conversations, and multiple language models to determine the robustness and generalizability of the observed behavior. Future evaluation will focus on quantitative precision, recall, F1, false-positive and false-negative rates, together with confidence intervals, to establish statistically stronger evidence for practical deployment.

PROJECT SUPERVISOR

PROJECT CO-ORDINATOR

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AD23722– Project Phase-1 (2023– 2027 BATCH)
PROJECT PHASE-I STUDENT CALENDAR
IV YEAR PROJECT PHASE I- REVIEW REPORT-(2026-27)

TITLE	Self-Refine Guardrail (SRG): A Two-Gate Framework for Automated Response Safety and Quality Verification
TEAM MEMBERS REG. NO & NAME	Jashareen J (231801066) - Meenakshi S S (231801100) – Vidhursh Kumar (231801186)
PROJECT MENTOR	
REVIEW	ZEROTH/FIRST/SECOND/THIRD/FOURTH
REVIEW DATE	
TEAM NUMBER	2027_PR_61

Suggestion and Comments for the Progress Required: (TO BE FILLED BY SUPERVISOR AND CO-ORDINATOR)

SUPERVISOR	
COORDINATOR	

Measures Taken for Progress: (TO BE FILLED BY STUDENT)

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PROJECT SUPERVISOR

PROJECT CO-ORDINATOR